Virtual Labs Experiment (Integrator and Differentiator)

Differentiator

Aim

To study the behavior of an operational amplifier (op-amp) circuit configured as an integrator and differentiator using resistive and capacitive elements, and to analyze the output response for sine and square wave inputs.

Theory

Operational amplifiers are versatile components used in analog signal processing. By appropriately placing resistors and capacitors in the input and feedback paths, op-amps can perform mathematical operations:

- **Integrator:** A resistor at the input and capacitor in the feedback path. The output is the time integral of the input signal.
- **Differentiator:** A capacitor at the input and resistor in the feedback path. The output is the time derivative of the input signal.

For sinusoidal inputs, the integrator introduces a phase lag of 90°, while the differentiator introduces a phase lead of 90°. For square wave inputs, the integrator produces a triangular waveform, while the differentiator produces sharp spikes at transitions.

Procedure

1. Configure the virtual lab circuit with the op-amp, resistor, capacitor, and input source.
2. Select the input waveform (sine or square) from the source.
3. Connect the output (Vo) to the oscilloscope.
4. Observe the input and output simultaneously using Ch1 and Ch2.
5. Record the waveforms for both integrator and differentiator configurations.

Results

- **Square Wave Input:**
  - Integrator: Output is triangular.

- Differentiator: Output shows sharp pulses at rising and falling edges.

- **Sine Wave Input:**
  - Integrator: Output lags by 90° (cosine waveform).
  - Differentiator: Output leads by 90° (cosine waveform).

Conclusion

The experiment demonstrates how op-amp circuits can be configured to perform integration and differentiation of input signals. The integrator converts signals into their time integrals, while the differentiator produces their time derivatives. These fundamental operations are widely applied in signal processing, control systems, and analog computation.

Simulation images:

Differentiator:

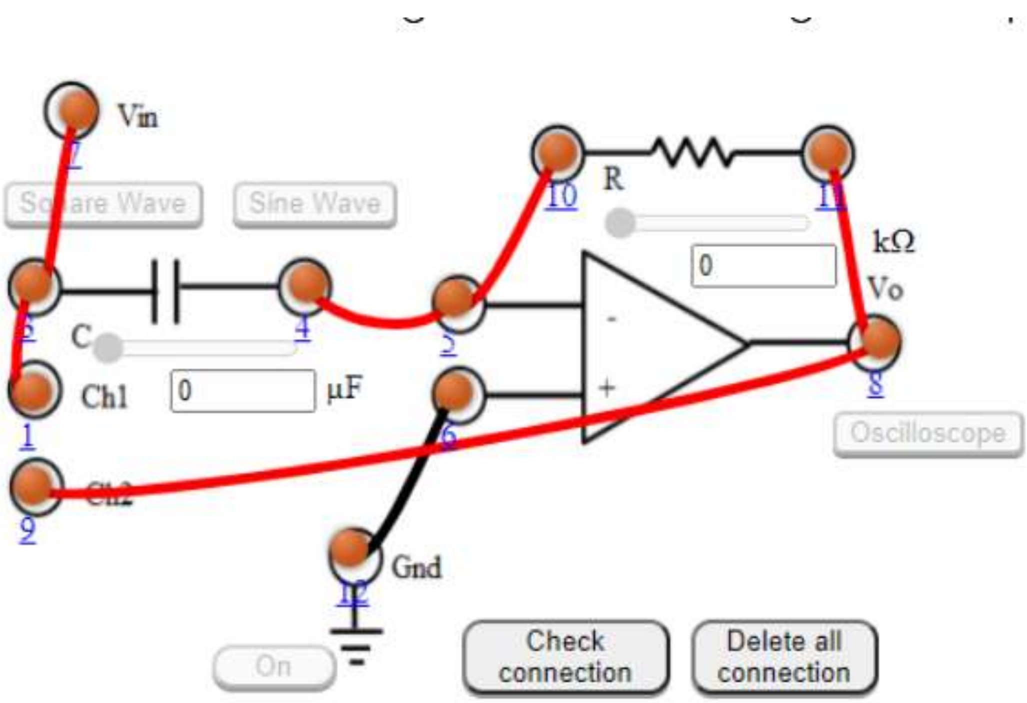
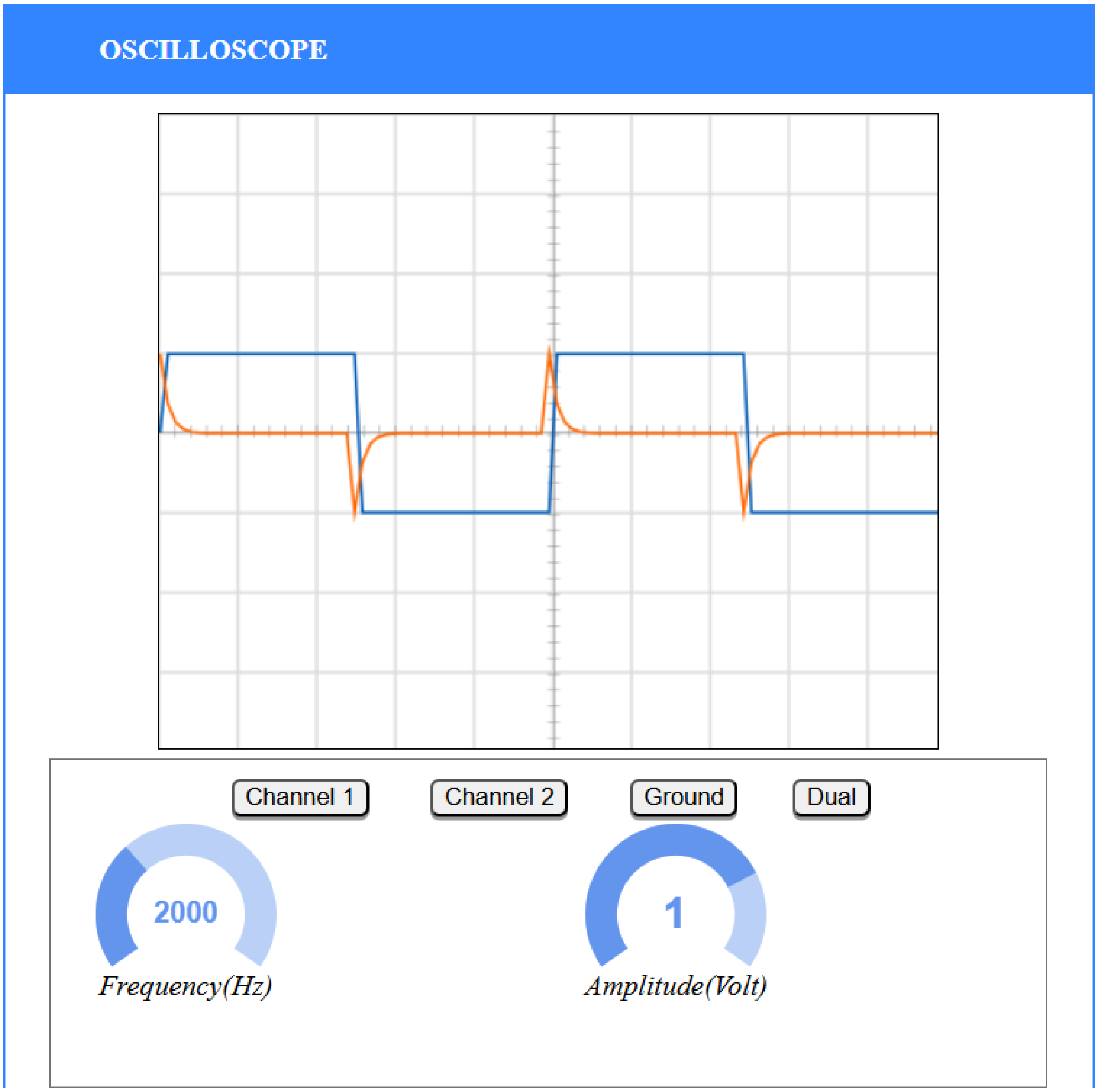


Figure: 2



INTEGRATOR:

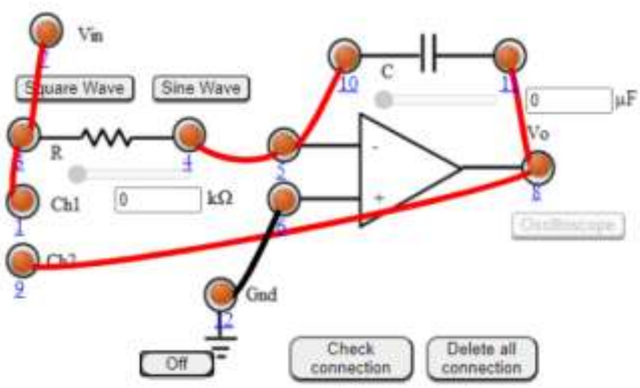
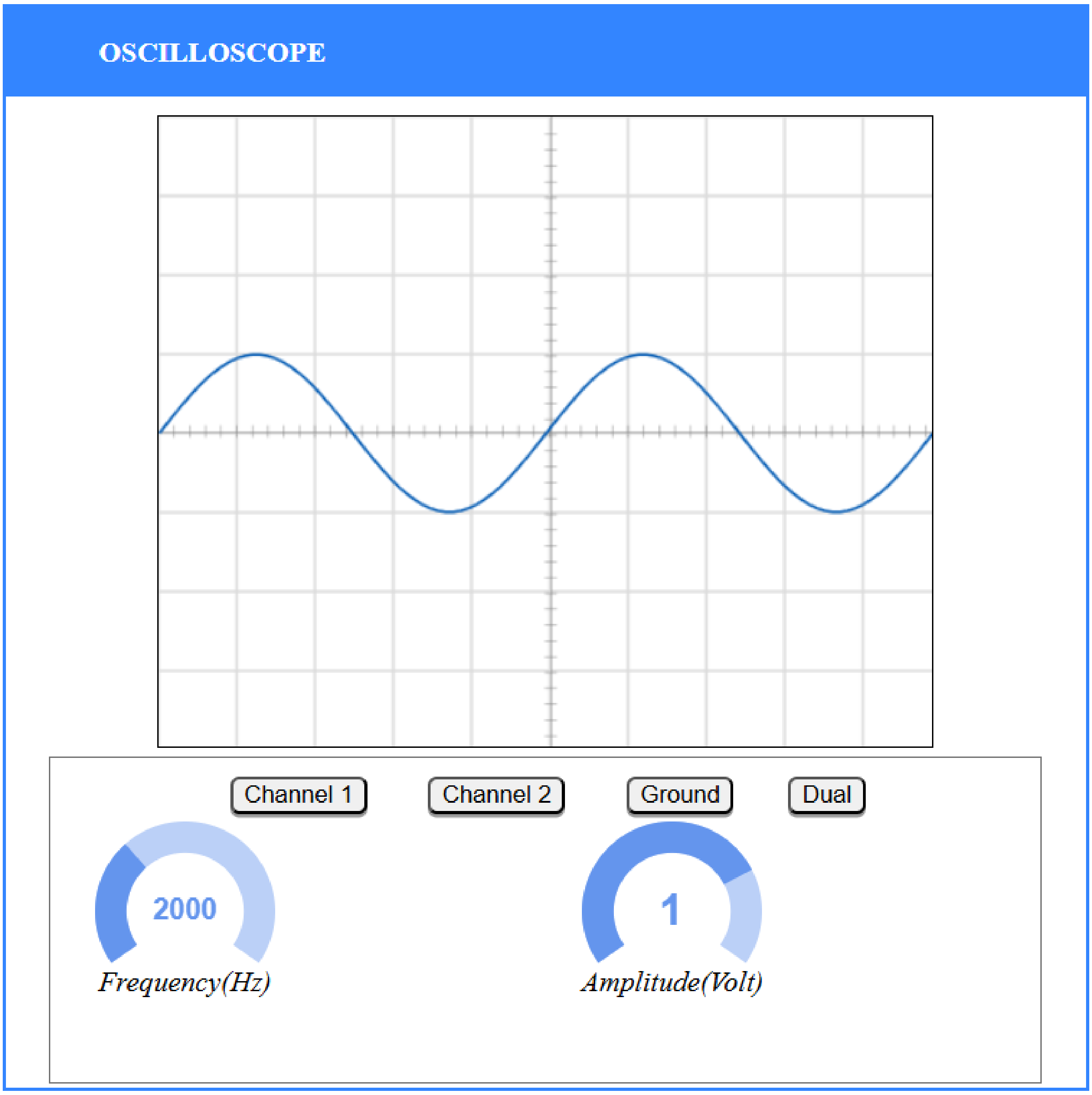
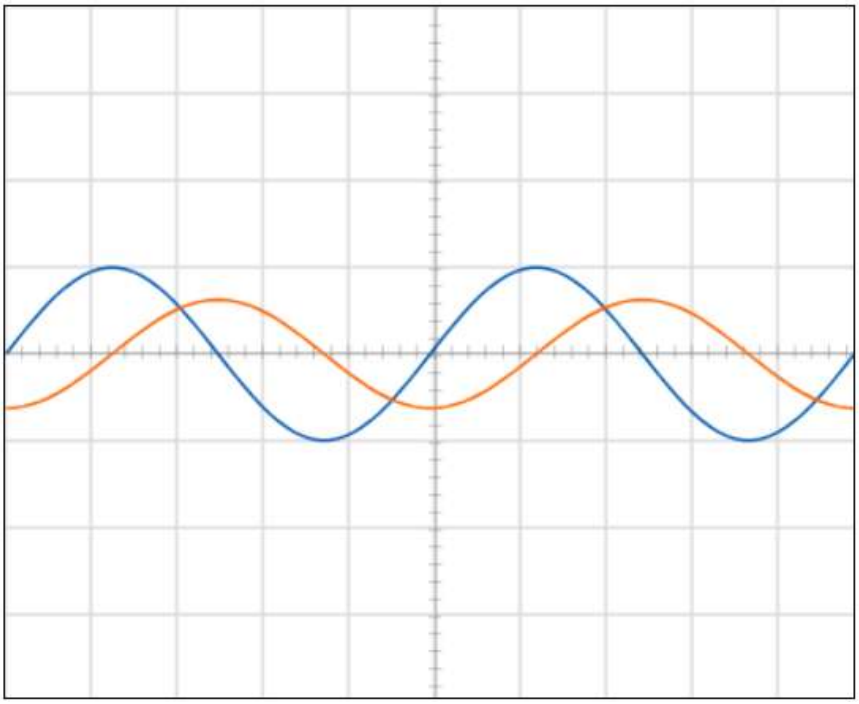


Figure:1


OSCILLOSCOPE
Channel 1
Channel 2
Ground
Dual
2000
Frequency(Hz)
1
Amplitude(Volt)

INSTRUCTION

OSCILLOSCOPE



Integrator


OSCILLOSCOPE
Channel 1
Channel 2
Ground
Dual
1000
Frequency(Hz)
1
Amplitude(Volt)


OSCILLOSCOPE
Channel 1
Channel 2
Ground
Dual
1000
Frequency(Hz)
1
Amplitude(Volt)